Universität St.Gallen

Using Computer Simulation

for Understanding Organizations as Complex Systems

Week 2: How to apply Agent-Based Modelling?

Marcel

Structure

Date	Topic	Deliverables	Literature	Responsible
19.02.	Introduction to the course, theory foundations, simulation demos	-	papers 1, 2	Stephan & Marcel
26.02.	**How to apply agent-based modelling?**	-	book chapter 1	Marcel
05.03.	How to implement simulation models with NetLogo?	-	book chapter 2 (3, 4, 5)	Marcel
12.03.	Independent case work (*no input session*)	-		-
19.03.	Intermediate group presentation and feedback on case/model	Presentation		Stephan & Marcel
26.03.	How to run simulation experiments and analyse simulation data?	-	book chapter 6	Stephan
	(*break*)			
16.04.	Intermediate group presentation on implemented model and analysis	Presentation		Stephan & Marcel
23.04.	How to verify and validate simulation models?	-	book chapter 7	Stephan
30.04.	Independent case work (*no input session*)	-		-
07.05.	Final case presentation in groups and reflection of learnings	Presentation, Model, Analysis		Stephan & Marcel
14.05.	Ascension Day (*no input session*)	-		-
21.05.	Oral exams	Oral Exam		Stephan & Marcel

Paper 1: Beese, J., Haki, M. K., Aier, S., & Winter, R. (2019). Simulation-Based Research in Information Systems: Epistemic Implications and a Review of the Status Quo. Business & Information Systems Engineering, 61(4), 503-521.
Paper 2: Haki, K., Beese, J., Aier, S., & Winter, R. (2020). The Evolution of Information Systems Architecture: An Agent-Based Simulation Model. MIS Quarterly, 44(1), 155-184.
Book: Wilensky, U., & Rand, W. (2015). An Introduction to Agent-Based Modeling: Modeling Natural, Social, and Engineered Complex Systems with NetLogo. The MIT Press.

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Using Computer Simulation for Understanding Organisations as Complex Systems

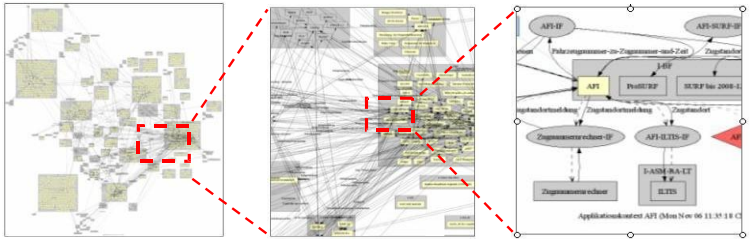
Using Computer Simulation for Understanding Organisations as Complex Systems

Core concepts to remember:

- Complex Systems:
  - Many and diverse objects
  - Many dependencies (structural & dynamic)
  - Many changes (system has a "behaviour")
- Complex Adaptive Systems (CAS):
  - CAS are complex systems (above) that show non-linear system behaviour as a result of the adaptive nature of its elements. The system behaviour emerges from interactions within the system.
- Emergence:
  - Emergence is the arising of novel and coherent structures, patterns, and properties through the interactions of multiple distributed elements.

"Complex systems theory (…) defines complex systems as systems that are composed of multiple individual elements that interact with each other yet whose aggregate properties or behaviour is not predictable from the elements themselves." (p.6)

Wilensky (2015)



Notice the levels of abstraction between individual actions/actors and the system-level behaviour.

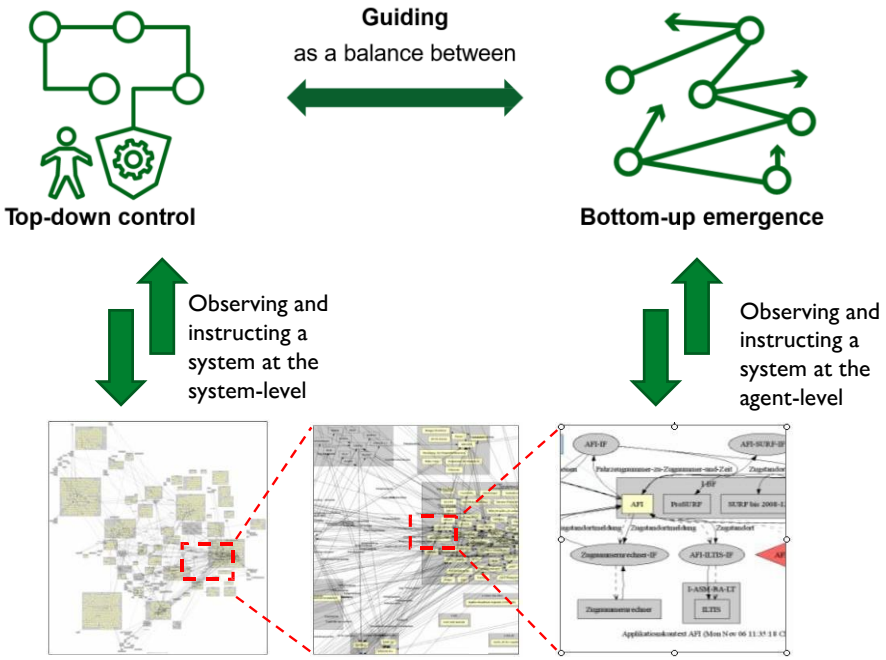
Using Computer Simulation for Understanding Organisations as Complex Systems

We can **observe, describe, and criticise** an organization's behaviour … but where does it come from? **(How) Can we change it?**

The understanding of management can be split into **two extreme forms**:

- **Top-down control:** Decisions and behaviour depend on prescriptive rules that need to be defined. The overall system behaviour can controlled.
- **Bottom-up emergence:** Decisions and behaviour are the result of individual's decision architectures. The overall system behaviour emerges.

Management is guiding an organisation's behaviour by balancing controls and by shaping emergence.



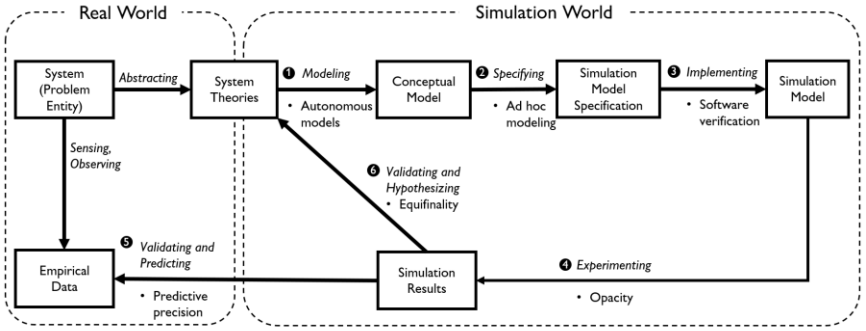
Notice the levels of abstraction between individual actions/actors and the system-level behaviour.

Using Computer Simulation for Understanding Organisations as Complex Systems

An organisation's behaviour can be guided by both controls and by shaping emergence—but how can we predict its behaviour (given the difficulty of anticipating system-level emergence)?

We may use research, including:

- Existing theory (find explanations that could fit)
- Existing empirical data ((re-)analyse data)
- Generate new data
  - Surveys
  - Interviews
  - Observation studies
  - …
  - **Computer Simulations !**



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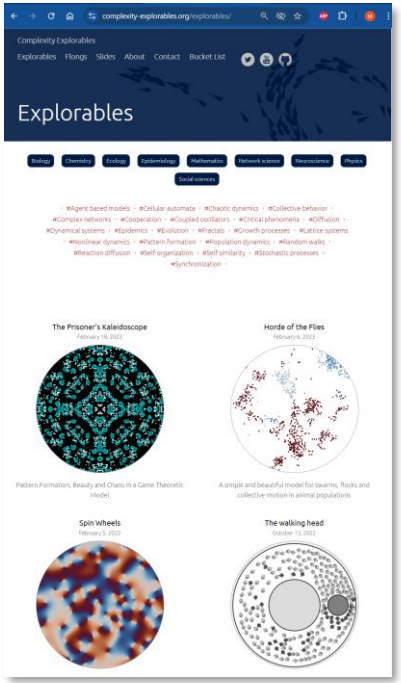
By generating data with computer simulations, we can avoid adverse effects of practically testing (potentially bad) ideas, overcome lacking data/access to data, compare many different scenarios, accelerate data collection compared to real-world field experiments, and much more.

Phenomena and how they can be presented in simulations

We discussed complex systems implemented in computer simulations (complexity-explorables.org):

1. What is being modeled?
2. What are the model's components?
3. What are your configuration options? (IVs)
4. What can you observe? (emergent behaviour)
5. How could the model behaviour be mapped to an organization/management?



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What is a Model?

Examples:

Definitions:

What is a Model?

According to the General Model Theory (Allgemeine Modelltheorie, Stachowiak, 1973, p.131-133):

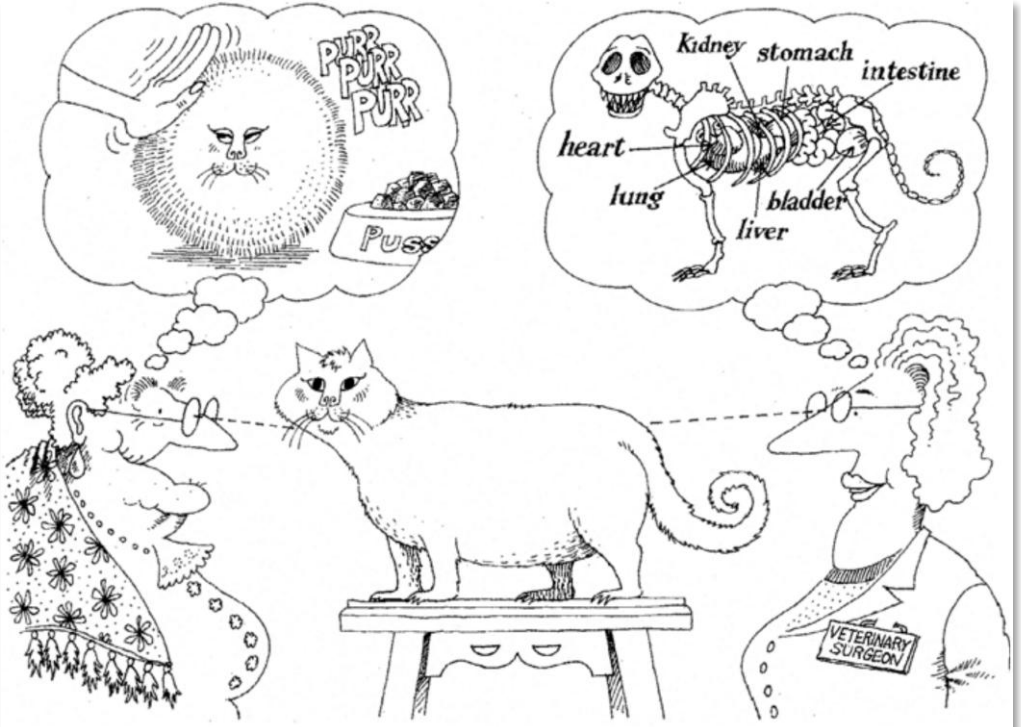
A model is a **representation of an original**.

A model has three essential characteristics. A model is:

- a **representation**: A model's constituents map to the original's characteristics (but they are not identical).
- **reductionist**: A model only contains the characteristics of the original that appear relevant to the modeller.
- **pragmatic**: A model is created for a specific user, in a certain context and with a purpose.*

*Multiple models may represent the same phenomenon in a reductionist way but differ strongly from each other. To use a model, one should be aware of the specific model's target audience (for whom?), its creation purpose (for what?) and its context (e.g., in which time period was it created?).

What is a Model?



Abstraction focuses on the essential characteristics of some object, relative to the perspective of the viewer.

Source: Booch, G. et al. (2007). *Object-Oriented Analysis and Design with Applications* (third edition). Addison-Weslley. p.45.

What is a Model?

Hence, models might be a good starting point for "understanding organisations as complex systems."

From our course literature:

"A model is **descriptive of a real-world system** … (A)ll models are **coarse-grained descriptions of reality**; (…) models that are not coarse-grained descriptions are useless as descriptions because they are indistinguishable from the real world and therefore do not assist in our understanding of complex systems (Korzybski, 1990). If your model includes all aspects of the real phenomenon, it is more efficient to simply observe reality (…). **The function of a model is to help us to understand and examine phenomena** that exist in the real world **in more tractable and efficient ways than by simply observing** reality." (Wilensky & Rand, p. 28-29)

"Models are explanatory in that they point out the **essential mechanisms** underlying a phenomenon. They can function as a proof that hypothesized mechanisms are sufficient to account for an observation." (Wilensky & Rand, p. 29)

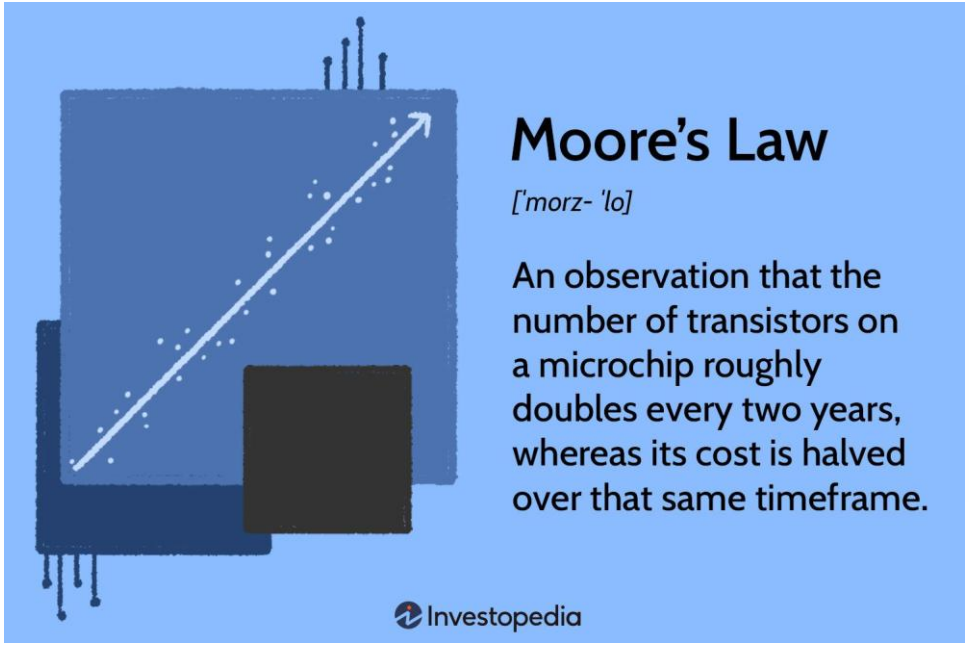
- A model is a coarse-grained representation of a phenomenon
- that features its essential underlying components and mechanisms.
- It helps understand the phenomenon,
- and it makes its understanding more efficient than observing it in reality.

Discussing models with this understanding

Let's look at some models and check, whether our definition(s) apply:

A model has three essential characteristics:

- a **representation**
- **reductionist**
- **pragmatic**



Modelling: Creating conceptual models

What is a model? (see above)

Three essential characteristics (Stachowiak, 1973):

- a **representation**
- **reductionist**
- **pragmatic**

A model is (Wilensky & Rand):

- a **coarse-grained** representation of a phenomenon
- features its **essential components** and **mechanisms**
- **helps understand** the phenomenon

… and (thus) makes understanding more efficient than observing the phenomenon in reality.

What does this mean for modelling?

Modelling is the process one goes through between identifying phenomenon and having a reductionist, pragmatic representation of that phenomenon.

1. Identify a phenomenon
2. Narrow down its scope:
   a) What is the use for your model?
   b) Which parts of the phenomenon are **essential** to your use case; and which parts are not?
3. What are essential components and mechanisms?
4. Represent the essential components and mechanisms in a way that helps understanding.

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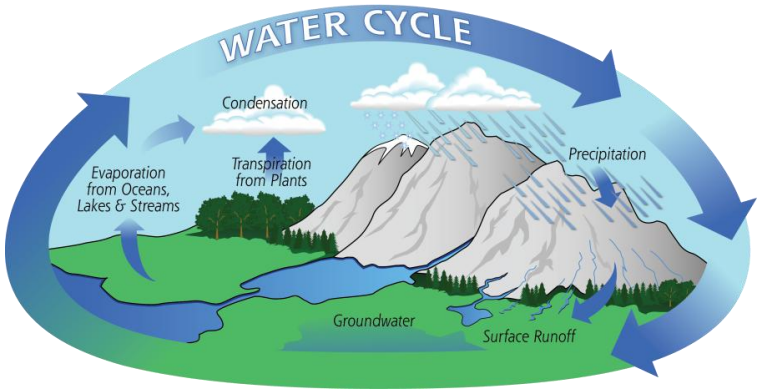
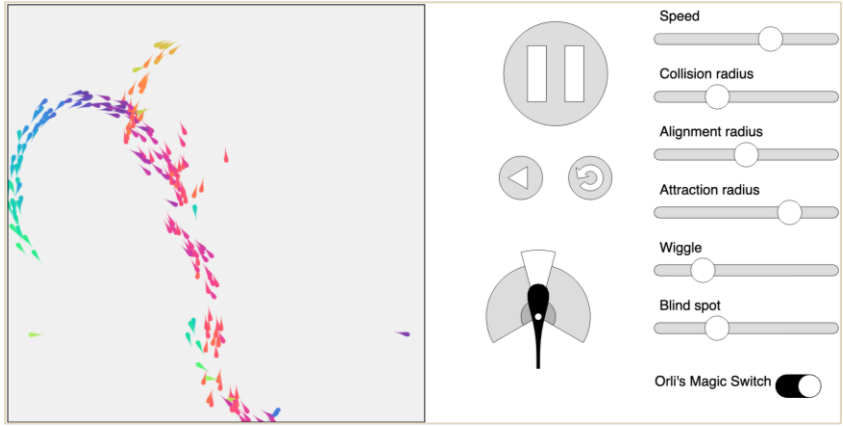
What is ABM?



Agent:
Agent-based:
Modelling:

Wilensky & Rand:
"Agent-based modelling is a form of computational modelling whereby a phenomenon is **modeled in terms of agents and their interactions**." (p.1)



Source: complexity-explorables.org

*Wilensky & Rand: "An agent is an autonomous computational individual or object with particular **properties and action**." (p.1)

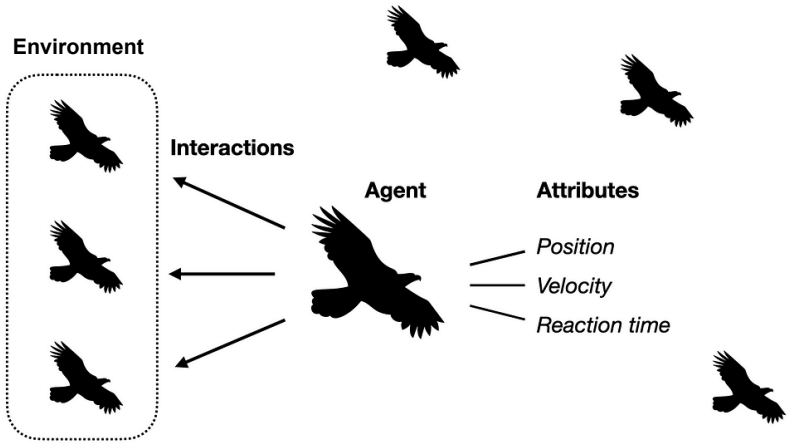
What is ABM?

Agent-based Models consist of
- **Agents** (one or several types of agents; incl. attributes)
- **Environment** (space or context for agents' actions)
- **Interactions** (update scheme for the attributes)

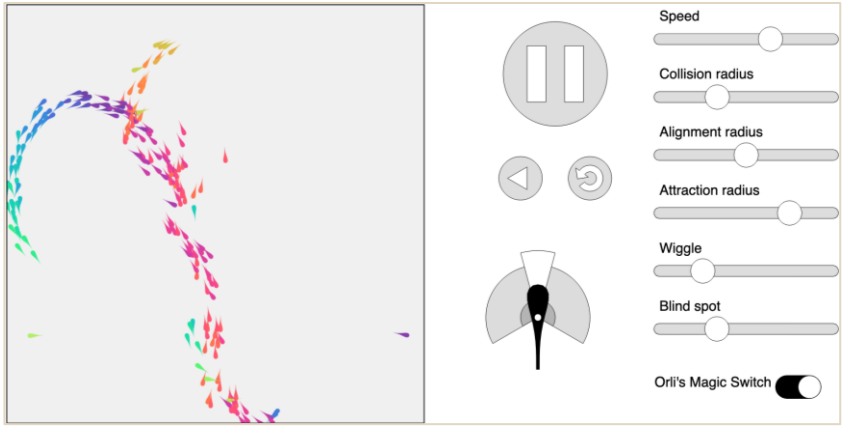
Agents are described with **relevant attributes** and fundamental behavioural rules.

The **environment** can be spatial or non-spatial, dynamic or static. It may thus limit or enable interaction and it may change (and thus have its own attributes) or not change over time.

Interactions consist of the interaction mechanisms (e.g., local vs. global interaction; direct vs. indirect interaction) and interaction rules (e.g., interaction protocols, strength modifiers) that govern the interactions between agents and between agents and the environment.



Source: towarddatascience.com



Source: complexity-explorables.org

Why use ABM?

"Using computer simulation for understanding organisations as complex systems." → We use ABM. Why?

- Agent-based Modelling is most useful when the task is to capture individual behaviours, local interactions, and the **emergence of complex dynamics from simple rules**.
- It is great for creating a better **understanding of complex systems** (dynamic, heterogeneous, multi-actor, …), because it captures that complexity through agent-agent and agent-environment interaction.
- ABM allows for studying social and socio-technical systems by **including messiness** (stochasticity to help mimic the randomness in actor behaviour more accurately)
- ABM can be used to both study the **status quo** (a real-world phenomenon) **and hypothetical phenomena** (by altering parameters)

- Agent-based Models are intuitive and (should be) easy to understand compared to e.g., mathematical models

Limitations of ABM?

Theoretically, we could model any phenomenon that includes agents (components with agency) with ABM.

When not to use ABM?

- Homogenous agents → use aggregating approaches, e.g., system dynamics, since agent-level is less/not relevant
- Heterogeneity of agents does not affect system behaviour → no need to track agent-level behaviour
- Excessively large number of agents → simplify and use aggregation (prohibitive computational requirements)

Criticisms and limitations:

- **Emergence is hard to explain:** Simulation results may be easy to capture visually, but hard to explain.
- **Analytical rigor**: Emergence can be serendipitously discovered rather than carefully and deterministically derived. The complexity and stochastic nature of ABMs make it hard to ensure convergence.
- **Parameter sensitivity and calibration**: ABS can be highy sensitive to initial parameters and may require extensive calibration for them to produce useful output. This fine-tuning is sometimes perceived as arbitrary.
- **Complexity and Overfitting**: ABM quickly become excessively complex. Calibration may lead to realistic system behaviour—but insights and models might not be transferrable (i.e., mapped to other contexts).
- **Scalability:** With increasing n-agents, computational requirements increase (compared to other models)

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How to apply ABM?

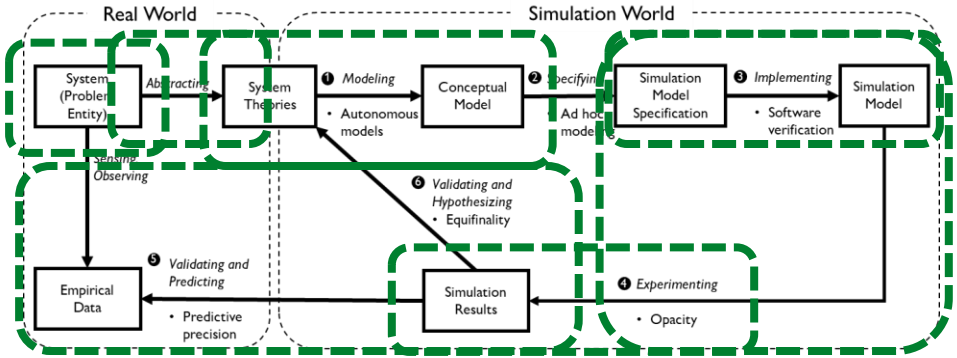
Approach

1. Identify a phenomenon and formulate a question

2. Narrow down the scope of the phenomenon

3. Identify and define actors (incl. relevant attributes and behaviours), environment (incl. non-spatial vs. spatial, dynamic vs. constant), & interactions (including stochasticity in actor, env., interactions).

4. Implement the model (NetLogo, week 3)
5. Calibrate the model (week 4)
6. Run simulations (week 6)
7. Validate the model & draw conclusions (week 9)



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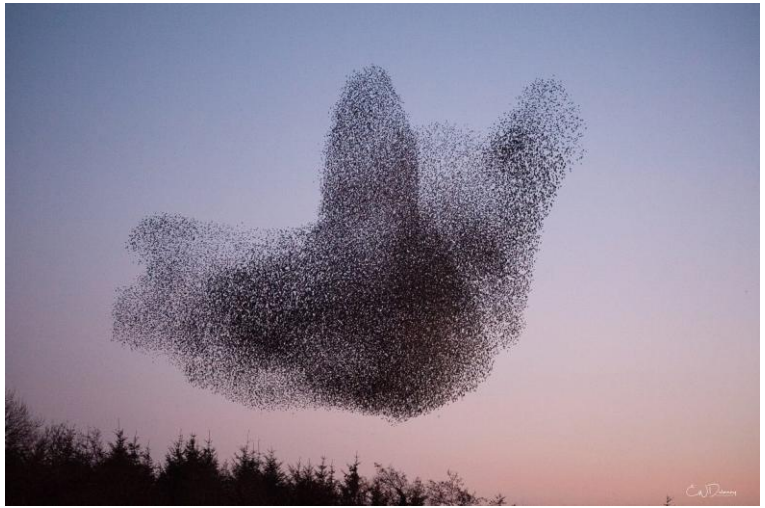
Example

1. Birds flock and create spectacular moving shapes. How do they coordinate their behaviour?



Source: BirdWatchIreland, birdwatchireland.ie

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Example

2. Only care about starlings' flight pattern and how they coordinate their individual behaviour.
   - Within scope: individual movement of birds of the species starling (speed, direction, awareness of others and such), aggreageted bird movement (overall shape and behaviour of the flock)
   - Out of scope: all other variables, such as weather (e.g., temperature, wind, wind direction, time of day) or the physical environment (e.g., trees, houses, windmills), all behavioural patterns that are not related to the secies starling (e.g., V-shape flight of geese)

How to apply ABM?

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Example

3. Actors: birds
   attributes: speed (s), angle (a), wiggle (da), vision radius (vr), collision radius (cr);
   behaviours: separate() if too close, align() if unaligned, cohere() if distanced, fly() at speed s in direction of angle a + wiggle ([-da - da])

   Environment: constant, spatially limited 2D map.

   Interaction:
   1. Identify birds within radius vr for each bird
   2. If >0 birds within vr,
      check distance to the closest bird per bird
      If distance < cr
         separate()
      otherwise: align() and cohere()

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Assignment

1. Identify 2-3 interesting, non-trivial **organisational** phenomena.
   Specify a question about each of them (individual work, 15 min).

2. In groups of 4. Discuss the phenomena & questions. Agree on one. (15 min)

3. Narrow down the phenomenon's scope
   (What is relevant or not to your question? 10 min)

4. Identify and define actors (incl. relevant attributes and behaviours), environment (incl. non-spatial vs. spatial, dynamic vs. constant), & interactions. (20 min)

5. Prepare for presenting your results (What is the phenomenon? What is your view-point/question? Who/What are the actors? How do they interact?, 5 min)

6. Presentations and discussion of results (25 min)